

WEEK 1 TASK

Developing a Data Exploration Strategy Roadmap

Python-Based Exploratory Data Analysis Project

Project Theme	Exploring factors affecting passenger survival
Proposed Dataset	Titanic passenger dataset (publicly available dataset)
Primary Language	Python
Project Type	Data Exploration and Exploratory Data Analysis (EDA)

1. Introduction

Data exploration is an essential stage of a data science project because it helps transform raw data into meaningful information. Before building predictive models, a data scientist must understand the structure, quality, distribution, relationships, and limitations of the available data. This project proposes a structured Python-based exploration of the Titanic passenger dataset.

The main problem is to understand which passenger characteristics were associated with survival. The analysis will examine variables such as passenger class, sex, age, fare, family relationships, and embarkation location. The project is designed as a roadmap rather than a final predictive model, so the focus is on planning a reliable and repeatable exploration process.

2. Problem Definition

The Titanic dataset contains passenger-level information and a survival outcome. The project will investigate patterns in the data and identify variables that appear to have meaningful relationships with survival.

Problem statement:

- How can systematic data exploration using Python identify important patterns, relationships, data-quality issues, and potential factors associated with passenger survival?

3. Project Objectives

- Understand the structure, columns, data types, and basic characteristics of the dataset.
- Identify missing values, duplicate records, inconsistent values, and potential outliers.
- Summarize important numerical and categorical variables using descriptive statistics.
- Study relationships between passenger characteristics and survival.
- Create clear visualizations to communicate major patterns and differences.
- Form and test exploratory hypotheses using appropriate statistical summaries and comparisons.
- Document assumptions, limitations, risks, and decisions so the analysis is reproducible.
- Prepare a clean analytical dataset and a set of findings that could support future predictive modeling.

4. Key Questions

1. Does passenger sex have a strong relationship with survival?

2. Did passenger class influence survival probability?
3. How did age relate to survival?
4. Was survival different for passengers traveling alone versus with family?
5. Did fare level show an association with survival?
6. Were survival rates different across embarkation locations?
7. Which variables contain substantial missing data and how might this affect conclusions?
8. Which variables appear most useful for a future machine-learning model?

5. Hypotheses

ID	Hypothesis	Exploration Method
H1	Female passengers had a higher survival rate than male passengers.	Compare survival proportions by sex.
H2	First-class passengers had a higher survival rate than second- and third-class passengers.	Compare survival proportions across Pclass.
H3	Younger passengers, particularly children, had different survival patterns from older passengers.	Group age into meaningful ranges and compare survival rates.
H4	Passengers traveling with immediate family members had different survival rates from passengers traveling alone.	Create family-size/travel-alone features and compare groups.
H5	Higher fare values were associated with higher survival rates, partly because fare is related to passenger class.	Compare fare distributions and survival across fare groups/classes.
H6	Embarkation location was associated with survival, although this relationship may be influenced by other passenger characteristics.	Compare survival rates by embarkation port and inspect class composition.

6. Data Sourcing Strategy

The initial dataset will be a publicly available Titanic passenger dataset. Public data is suitable for this project because it allows practical EDA without requiring private or confidential information.

Data acquisition process:

9. Identify a reputable public dataset repository such as Kaggle, UCI, or another established open-data source.
10. Verify the dataset description, column definitions, file format, licensing/usage conditions, and source information.
11. Download the CSV dataset and preserve the original file as a read-only raw-data copy.
12. Record the dataset source, access date, version if available, and any known limitations.
13. Load the dataset into Python and compare the observed columns with the published data description.
14. Create a separate cleaned/processed dataset rather than modifying the raw source.

7. Initial Tools and Technology Selection

Tool / Library	Purpose	Reason for Selection
Python	Main programming language	Widely used for data science and supports the complete EDA workflow.
Pandas	Data loading, cleaning, transformation, grouping	Provides efficient DataFrame operations and readable syntax.
NumPy	Numerical operations	Supports arrays, calculations, and numerical preprocessing.
Matplotlib	Basic charts	Flexible and suitable for creating publication-ready visualizations.
Seaborn	Statistical visualization	Makes relationships and group comparisons easier to visualize.
Jupyter Notebook / VS Code	Development and documentation	Allows iterative analysis, code execution, notes, and visualization.
CSV	Primary data format	Simple, portable, and directly supported by Pandas.
Git/GitHub	Version control	Helps track changes and maintain project history.

8. Methodology and Analysis Phases

Phase 1 – Problem Understanding

- Define the business/research question.
- Identify target variable (Survived) and explanatory variables.
- Document assumptions and expected outcomes.

Phase 2 – Data Acquisition

- Download the public dataset.
- Verify source, structure, size, and documentation.
- Store raw data separately from processed data.

Phase 3 – Data Understanding

- Inspect rows, columns, data types, unique values, and summary statistics.
- Check the target distribution.
- Identify categorical and numerical variables.

Phase 4 – Data Quality Assessment

- Measure missing values.
- Check duplicate records and suspicious values.
- Inspect impossible or extreme values.
- Decide how each quality issue will be handled and document the reason.

Phase 5 – Data Preparation

- Create a cleaned working dataset.
- Handle missing values using justified methods.

- Convert data types where necessary.
- Create useful derived variables such as FamilySize and IsAlone.

Phase 6 – Univariate Analysis

- Analyze individual variables using counts, mean, median, standard deviation, and distributions.
- Use histograms, boxplots, and bar charts where appropriate.

Phase 7 – Bivariate and Multivariate Analysis

- Compare survival with sex, class, age, fare, family size, and embarkation.
- Use grouped summaries, cross-tabulations, and visualizations.
- Look for confounding patterns, such as the relationship between class and fare.

Phase 8 – Hypothesis Review

- Compare evidence against each hypothesis.
- Clearly distinguish observed association from causation.
- Record supporting, weak, or contradictory evidence.

Phase 9 – Findings and Communication

- Summarize major findings.
- Identify the most informative variables.
- Present charts with clear titles and labels.
- Document limitations and recommendations for future modeling.

9. Planned Exploratory Analysis

- Dataset dimensions: number of rows and columns.
- Data types: numerical, categorical, and binary variables.
- Missing-value percentage for each column.
- Descriptive statistics for age, fare, and family-related variables.
- Survival rate overall and by major categories.
- Distribution of passengers by passenger class and sex.
- Age distribution and survival by age group.
- Fare distribution and possible outliers.
- Correlation/association analysis for suitable numerical variables.
- Cross-tabulation of survival against sex, class, family status, and embarkation.

10. Data Cleaning Strategy

- Missing values: quantify first; use median, mode, a meaningful category, or row/column exclusion only when justified.
- Duplicates: identify exact duplicates and determine whether they represent true duplicate observations.
- Data types: convert columns to appropriate numeric or categorical representations.
- Outliers: inspect rather than automatically delete; determine whether they are valid extreme observations.
- Inconsistent categories: standardize spelling, capitalization, and labels when necessary.
- Feature engineering: create FamilySize and IsAlone to improve interpretation.

- Reproducibility: perform cleaning through Python code rather than manual spreadsheet edits.

11. Visualization Plan

- Bar charts for survival rates by sex and passenger class.
- Histogram for age distribution.
- Boxplot for fare and age distributions.
- Count plots for categorical variables.
- Grouped bar charts for survival across age groups and family status.
- Heatmap/cross-tab visualization for selected relationships.
- Scatter plots where numerical relationships are meaningful.
- All charts will include informative titles, axis labels, legends where necessary, and appropriate units.

12. Timeline and Milestones

Week / Stage	Activities	Milestone	Deliverable	Estimated Effort
Week 1	Problem definition, hypotheses, source identification, tool selection, roadmap	Strategy approved	This roadmap document	30–35 hours
Week 2	Data acquisition, inspection, quality profiling	Dataset validated	Data understanding notes	20 hours
Week 3	Cleaning and feature preparation	Clean dataset ready	Cleaning notebook + processed data	20 hours
Week 4	Univariate and bivariate EDA	Core patterns identified	EDA notebook + charts	25 hours
Week 5	Multivariate analysis and hypothesis review	Hypotheses evaluated	Analysis results	20 hours
Week 6	Final documentation, visualization refinement, review	Final EDA package	Final report + presentation-ready findings	15 hours

13. Resource Allocation

- Human resource: 1 data science student/analyst responsible for data collection, coding, analysis, documentation, and review.
- Computing resource: personal computer capable of running Python, Pandas, NumPy, Matplotlib, and Seaborn.
- Software resource: Python environment, VS Code or Jupyter Notebook, Git, and a browser for dataset research.
- Storage resource: separate folders for raw data, processed data, notebooks/scripts, visualizations, and documentation.
- Time resource: approximately 30–35 hours for the Week 1 strategic planning task, followed by approximately 100 additional hours for execution and refinement.

14. Iteration and Checkpoints

15. Checkpoint 1: Confirm that the problem statement and hypotheses are specific and measurable.
16. Checkpoint 2: Validate dataset source, structure, and licensing/usage information.
17. Checkpoint 3: Review data-quality findings before making cleaning decisions.
18. Checkpoint 4: Validate that visualizations accurately represent the data and are not misleading.
19. Checkpoint 5: Compare results against every hypothesis and record evidence.
20. Checkpoint 6: Peer/self-review of notebook, documentation, reproducibility, and conclusions.

15. Risk Analysis and Mitigation

Risk	Potential Impact	Likelihood	Mitigation Strategy
Missing or incomplete data	Biased or weaker conclusions	Medium	Profile missingness early; document it; use justified imputation or exclusion.
Poor data quality	Incorrect analysis	Medium	Validate ranges, categories, duplicates, and data types before analysis.
Selection bias	Findings may not generalize	Medium	Clearly state that the Titanic dataset represents a specific historical population.
Confounding variables	False interpretation of relationships	High	Use grouped/multivariate comparisons and avoid claiming causation.
Outliers	Distorted statistics	Medium	Inspect extreme observations and use robust summaries such as median where appropriate.
Overfitting conclusions	False patterns from random variation	Medium	Use multiple views of the data and keep conclusions proportional to evidence.
Tool/environment issues	Delays in execution	Low	Use a requirements list, stable environment, and version control.
Reproducibility problems	Difficulty repeating results	Low	Keep analysis in scripts/notebooks and preserve raw data separately.

16. Evaluation Criteria for the Project

- Clarity: The problem, objectives, questions, and hypotheses are easy to understand.
- Completeness: The roadmap covers sourcing, preparation, analysis, visualization, evaluation, and reporting.
- Technical suitability: Selected Python tools are appropriate for the planned tasks.
- Data quality awareness: Risks related to missing values, outliers, duplicates, and bias are considered.
- Milestone quality: Each stage has a measurable checkpoint or deliverable.
- Reproducibility: Data, code, assumptions, and decisions are documented.

- Communication: Findings and visualizations are clear, concise, and understandable.
- Professionalism: The final project is logically structured and suitable for academic/project submission.

17. Expected Final Outcomes

- A cleaned and documented Titanic dataset.
- A reproducible Python/Jupyter EDA notebook.
- Descriptive statistics and data-quality assessment.
- A collection of meaningful visualizations.
- Evidence-based conclusions for the proposed hypotheses.
- A list of important variables that may be considered in a future machine-learning model.
- A documented set of limitations and recommendations for further work.

18. Proposed Project Folder Structure

```
data-exploration-project/
├── data/
│   ├── raw/
│   └── processed/
├── notebooks/
│   └── titanic_eda.ipynb
├── scripts/
│   ├── data_cleaning.py
│   └── analysis.py
├── visualizations/
├── reports/
└── README.md
```

19. Conclusion

This roadmap provides a structured plan for completing a Python-based data exploration project. The proposed approach begins with a clearly defined problem and hypotheses, followed by responsible data sourcing, systematic data-quality assessment, cleaning, exploratory analysis, visualization, hypothesis review, and final communication. The roadmap also includes milestones, resource planning, iteration checkpoints, and risk mitigation. Following this strategy will make the project more organized, reproducible, and suitable for future extension into statistical analysis or machine-learning modeling.

20. Final Checklist

- ☐ Problem statement finalized
- ☐ Objectives and research questions documented
- ☐ Hypotheses defined
- ☐ Public dataset source verified
- ☐ Python libraries selected with rationale
- ☐ Data acquisition process planned
- ☐ Data cleaning strategy documented
- ☐ Analysis phases defined

- ☐ Visualization plan prepared
- ☐ Timeline and milestones established
- ☐ Resources allocated
- ☐ Risks and mitigation strategies identified
- ☐ Evaluation criteria defined
- ☐ Expected outcomes documented
- ☐ Reproducible folder structure planned

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